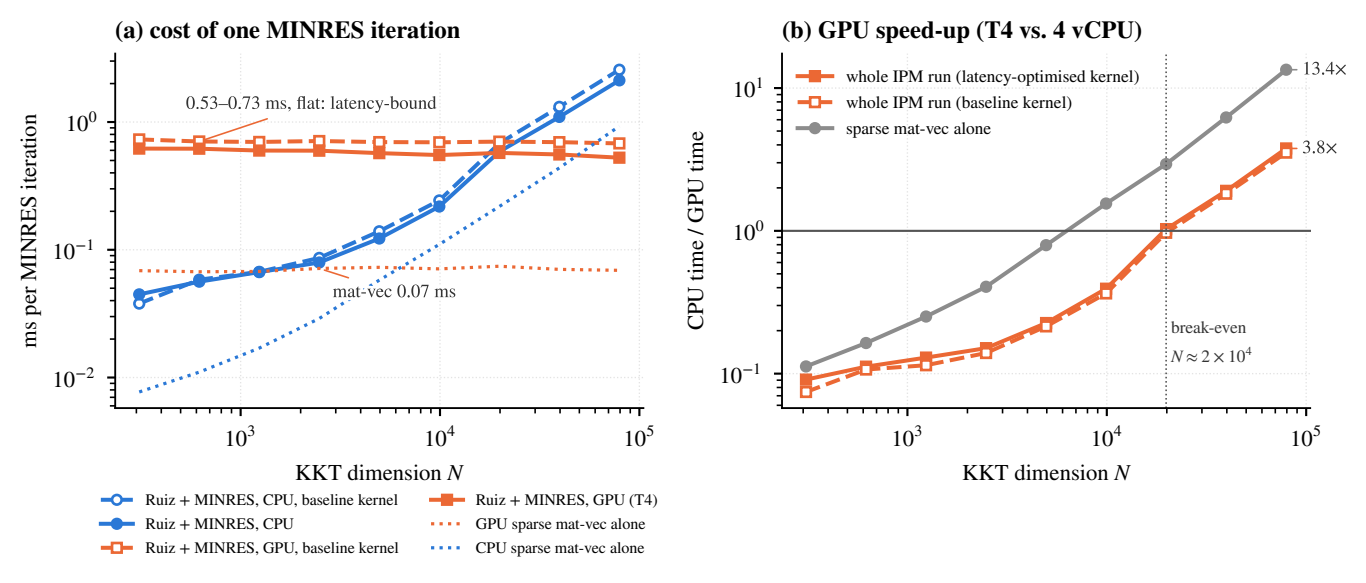

(a) cost of one MINRES iteration
ms per MINRES iteration
0.53–0.73 ms, flat: latency-bound
mat-vec 0.07 ms
KKT dimension N
Ruiz + MINRES, CPU, baseline kernel
Ruiz + MINRES, CPU
Ruiz + MINRES, GPU, baseline kernel
Ruiz + MINRES, GPU (T4)
GPU sparse mat-vec alone
CPU sparse mat-vec alone
(b) GPU speed-up (T4 vs. 4 vCPU)
CPU time / GPU time
whole IPM run (latency-optimised kernel)
whole IPM run (baseline kernel)
sparse mat-vec alone
13.4×
3.8×
break-even
$N \approx 2 \times 10^4$
KKT dimension N